Operations control room Specification

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Revision	C
Status	APPROVED
Classification	INTERNAL
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1. Purpose

This specification defines the performance, interface and verification requirements for the operations control room (OC-OPC) of the Operations Centre on Starhaven Orbital Habitat & Shipyard.

2. Scope

Fabrication and installation are performed under work package WP-SHV-OC-02 (WBS 1.17.2), owned by Wilde Ravell.

The prime supplier for flight hardware is Cygnus Spaceworks. Responsible engineer: Wilde Ravell.

3. Applicable documents

Document	Title
SHV-SYS-SPEC-0001	Starhaven Master System Specification
SHV-OC-DWG-0201	Operations control room general arrangement drawing
SHV-OC-BOM-0301	Operations control room bill of materials
SHV-QA-PLN-0004	Starhaven Verification and Test Plan
SHV-QA-STD-0002	Station Workmanship Standard
SHV-ICD-CN-OC	Interface Control Document: Computer Core & Data Network / Operations Centre

4. Description

The operations centre is where station operations, traffic control and facility management are run, with a conference suite and a shared operations display wall.

The operations control room is delivered as a set of line-replaceable assemblies, integrated on station by Construction Operations and verified by Quality Assurance & Test.

5. Performance requirements

ID	Requirement	Value	Tolerance	Verification
OC-OPC-R01	The operations control room shall provide a operator stations of 41 ea.	41 ea	±2%	Demonstration

5.1 General requirements

ID	Requirement	Verification
OC-OPC-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.	Inspection
OC-OPC-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.	Analysis
OC-OPC-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.	Inspection
OC-OPC-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.	Analysis
OC-OPC-G05	Documentation: Acceptance data packages shall be delivered with every unit.	Inspection

6. Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

7. Verification

Each requirement is verified by the method stated in section 5, following SHV-QA-PLN-0004. Acceptance test reports are filed under the OC test-report series.

8. Revision history

Rev	Date	Description	Author
A	2024-06-08	Initial release.	Wilde Ravell
B	2024-09-10	Updated requirement values after design review.	Wilde Ravell
C	2024-12-25	Clarified verification methods.	Wilde Ravell